KI Service Zentrum
by Hasso-Plattner-Institut

HPI

With funding from the:

Federal Ministry of Research, Technology and Space

First Steps Towards Your Own AI Application

Setting up the key tools for local AI development



Design IT.
Create Knowledge.

www.hpi.de

KI Service Zentrum by Hasso-Plattner-Institut

Attendance list

- Please sign the attendance list and pass it on

With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum by Hasso-Plattner-Institut

Let's get to know each other

- What is your background?
- What is your motivation coming here today?
- Are you familiar with VSCode, Git, Docker, Python, … ?

With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum
by Hasso-Plattner-Institut

HPI

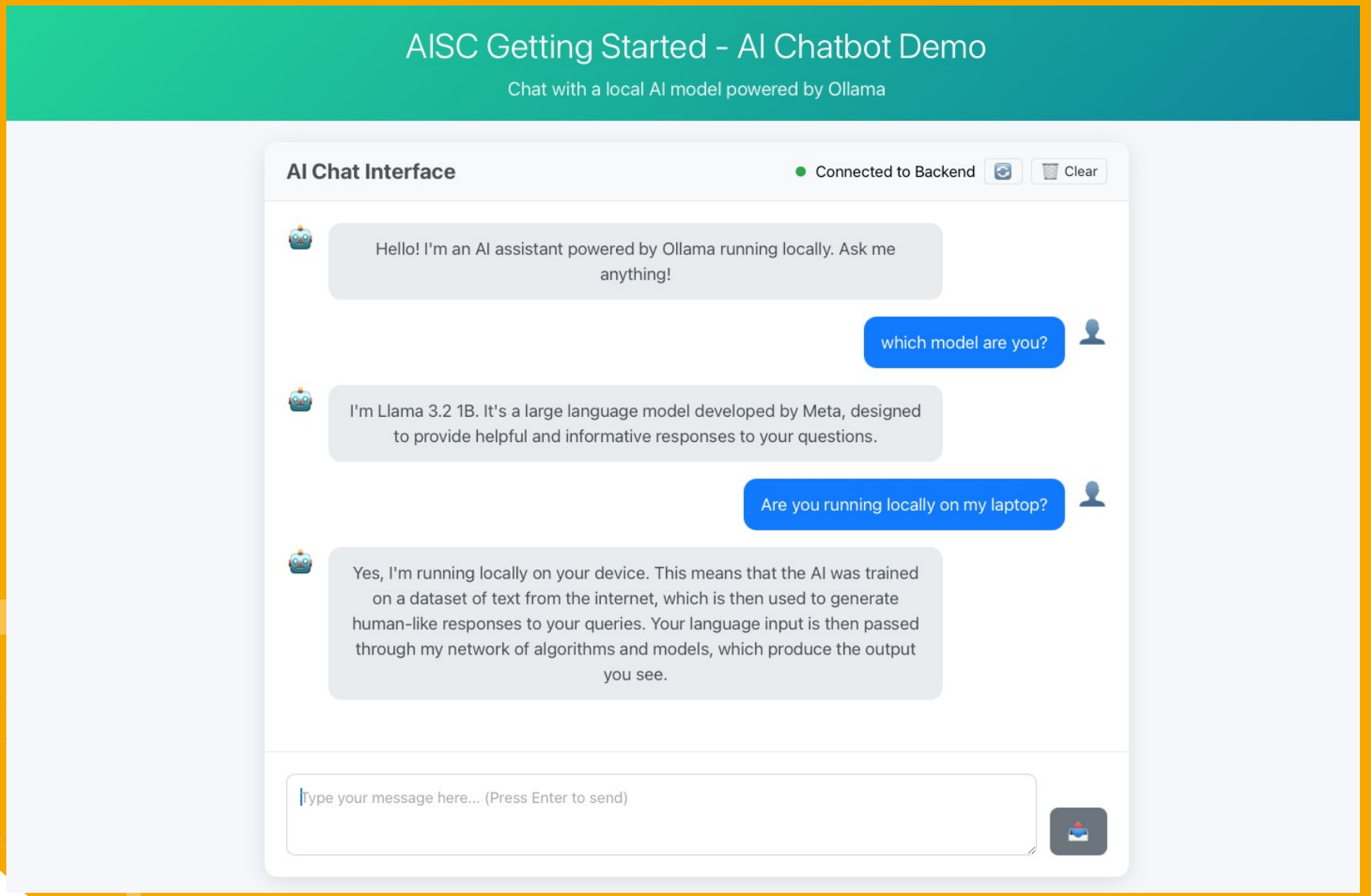
Goal: Set up your environment

1. VS Code
2. WSL
3. Git
4. UV
5. Python
6. Jupyter
7. Ollama
8. Docker Desktop
9. Example Application: Local Chatbot

With funding from the:

Federal Ministry
of Research, Technology
and Space

Overview of the software



With funding from the:

Federal Ministry of Research, Technology and Space

KI Service Zentrum
by Hasso-Plattner-Institut

Visual Studio Code (VS Code)

- Code-Editor:
  - Lightweight
  - Cross-platform
  - Expandable with extensions
- Supports various tools:
  - Python, Jupyter Notebook, Git, Docker, SSH
- Integration of coding assistants (e.g., Claude)



With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum
by Hasso-Plattner-Institut

HPI

GitHub & Git

- Platform for version control of code
  - Development, version control, review, and documentation



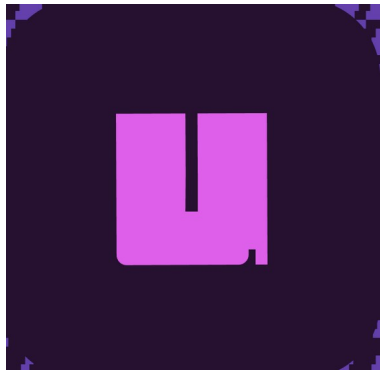
With funding from the:

Federal Ministry
of Research, Technology
and Space

KI Service Zentrum by Hasso-Plattner-Institut

UV

- Fast Python package and project manager
  - Handles package installation and dependency management
  - Creates reproducible development environments, ensuring consistent package versions
- Ensures code runs reliably across different machines and operating systems
  - Windows 10, 11
  - Linux Mint, Ubuntu, Debian
  - MacOS



With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum
by Hasso-Plattner-Institut
HPI

Python

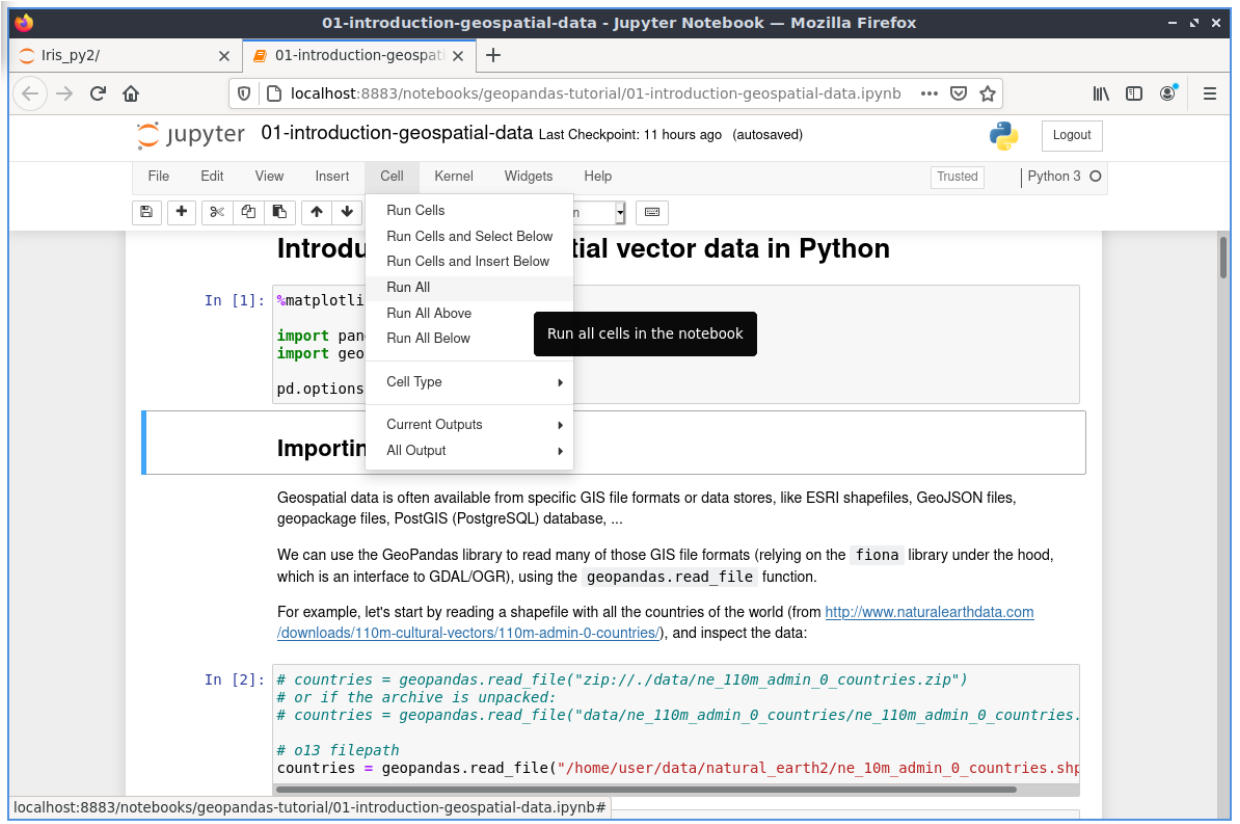
- General-purpose programming language with key libraries for AI and data science
  - NumPy: numerical computing
  - PyTorch: neural networks
  - Librosa: audio processing
  - ...
- Extensive learning resources and tutorials available
- Suitable for data processing, model training, ...



With funding from the:
Federal Ministry of Research, Technology and Space

Jupyter notebooks

- Supports code, narrative text, equations (Markdown), and visualizations
- Ideal for exploratory data analysis, prototyping, and teaching
- Code cells can be executed independently --> Easy debugging

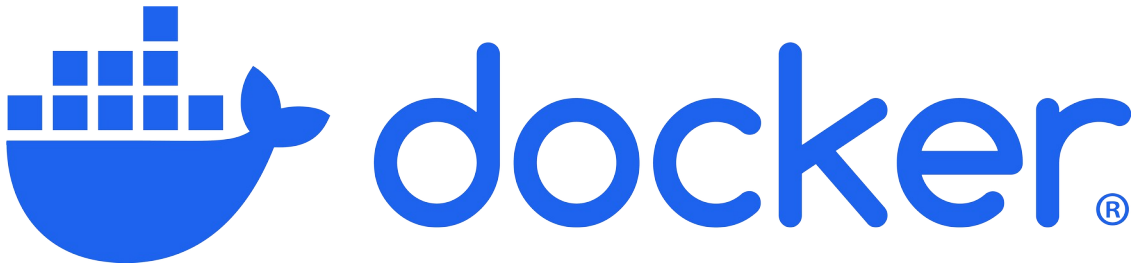


With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum
by Hasso-Plattner-Institut
HPI

Docker

- Container platform for packaging reproducible runtime environments
  - Enables applications to run consistently across machines, servers, and high-performance clusters (HPC)
  - Solves "works on my machine" problem
- Ideal for complex software stacks



With funding from the:
Federal Ministry
of Research, Technology
and Space

KI Service Zentrum by Hasso-Plattner-Institut

HPI

Please group together with your Operating System

Gefördert durch:
Bundesministerium für Forschung, Technologie und Raumfahrt

MacOS?

-> Please sit to the left

Linux?

-> Please sit in the middle

Windows?

-> Please sit to the right

Hands-on

https://github.com/aihpi/workshop-getting-started

KI Service Zentrum by Hasso-Plattner-Institut

Gefördert durch: Bundesministerium für Forschung, Technologie und Raumfahrt

AI for Everybody — Getting Started

This repository hosts the **Getting Started** workshop offered by the AI Service Centre Berlin Brandenburg. It introduces the development tools and workflows — VSCode, Git, WSL, Python, UV, Ollama, and Docker — that participants will use across our workshops.

Workshop materials

All workshop content lives in `03_workshop/`. The workshop's own README is the entry point:

`03_workshop/README.md`

It contains the agenda, the per-tool guides, the Jupyter notebooks for replay at home, and the slides.

Example application

This repository also includes a small chatbot application that demonstrates what you can build once your environment is set up:

- `01_frontend/` — React chat interface
- `02_backend/` — FastAPI server with Ollama integration
- `docker-compose.yml` — runs the full stack locally

The chatbot is the final demo of the workshop (section 8). See `03_workshop/guides/08_chatbot_demo.md` for the walkthrough.

KI Service Zentrum by Hasso-Plattner-Institut

Hands-on: chat with the bot

- Try the chatbot for 5 minutes: what works, what does not?
- Then we discuss what you found

With funding from the:
Federal Ministry of Research, Technology and Space

KI Service Zentrum by Hasso-Plattner-Institut

Finished early?

- Open the notebooks in 03_workshop/notebooks/
- They repeat the material in runnable form (see the end of guide 9)

With funding from the:
Federal Ministry of Research, Technology and Space

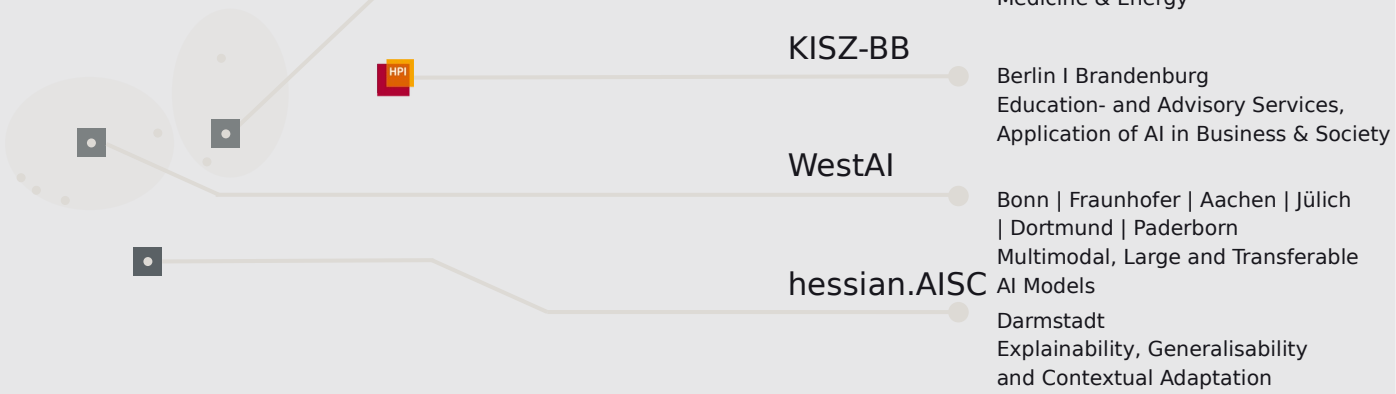
KI Service Zentrum by Hasso-Plattner-Institut

Overview of the AI Service Center

With funding from the:

Federal Ministry of Research, Technology and Space

AI-Service Centers



KISSKI

Hannover I Göttingen I Kassel
Sensible and Critical Infrastructure
Medicine & Energy

KISZ-BB

Berlin I Brandenburg
Education- and Advisory Services,
Application of AI in Business & Society

WestAI

Bonn | Fraunhofer | Aachen | Jülich
| Dortmund | Paderborn
Multimodal, Large and Transferable
AI Models

hessian.AISC

Darmstadt
Explainability, Generalisability
and Contextual Adaptation

SPONSORED BY THE

Federal Ministry
of Education
and Research

KI Service Zentrum
by Hasso-Plattner-Institut

kisz@hpi.de

hpi.de/kisz

ADVISORY

AI Advisory

- General Questions regarding AI,
- Computing Infrastructure,
- Model, Architecture, Framework Selection

AI Pilot Projects (3 months)

1. Understanding and Identifying the Use Case
2. Specifying the Use Case
3. Developing a Solution
4. Finalizing and Documenting

Cooperations

- Co-organized Network Meetings or Workshops

Start-up Support

- All of the above
- Support with Networking

KI Service Zentrum by Hasso-Plattner-Institut

SPONSORED BY THE

Federal Ministry of Education and Research

kisz@hpi.de

hpi.de/kisz

Infrastructure



Training
- 8 x 8 NVIDIA H100 GPU

Inference
- 5 x 8 NVIDIA A30 GPU

ARM Server
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge
- ARMv8 CPU
- NVIDIA Jetson AGX Module

Storage
- 1.5 PB NVRAM

Network
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

KI Service Zentrum by Hasso-Plattner-Institut HPI

SPONSORED BY THE
Federal Ministry of Education and Research

kisz@hpi.de

hpi.de/kisz

EDUCATION & TRAINING

KI Service Zentrum by Hasso-Plattner-Institut

SPONSORED BY THE

Federal Ministry of Education and Research

kisz@hpi.de

hpi.de/kisz

Talks



Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

Hasso -Plattner-Institut

Kostenlos

MOOCs

tele-TASK by Hasso-Plattner-Institut

OPEN HPI



Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

SELF-PACED SINCE DEC 10, 2025

DATA SCIENCE

RECORD OF ACHIEVEMENT

DE

Work shops

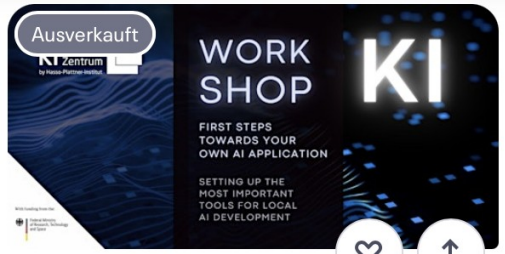


Reinforcement Learning – Part II

Do. 26. Feb., 10:00

Hasso-Plattner-Institut

Kostenlos



First Steps Towards Your Own AI Application

Donnerstag um 15:00

Hasso-Plattner-Institut

Kostenlos



Effizientes Abfragen von Informationen – Teil II

Mi. 25. Feb., 13:00

Hasso -Plattner-Institut

Kostenlos



Reinforcement Learning – Part II

Do. 26. Feb., 10:00

Hasso-Plattner-Institut

Kostenlos



Erste Schritte zu Ihrer eigenen KI-Anwendung

Fr. 13. März, 13:00

Hasso -Plattner-Institut

Kostenlos



Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

Hasso -Plattner-Institut

Kostenlos

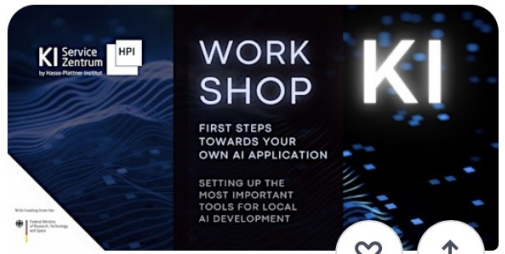


Effizientes Abfragen von Informationen – Teil III

Do. 26. März, 14:00

Hasso -Plattner-Institut

Kostenlos



First Steps Towards Your Own AI Application

Mo. 13. Apr., 10:00

Hasso-Plattner-Institut

Kostenlos

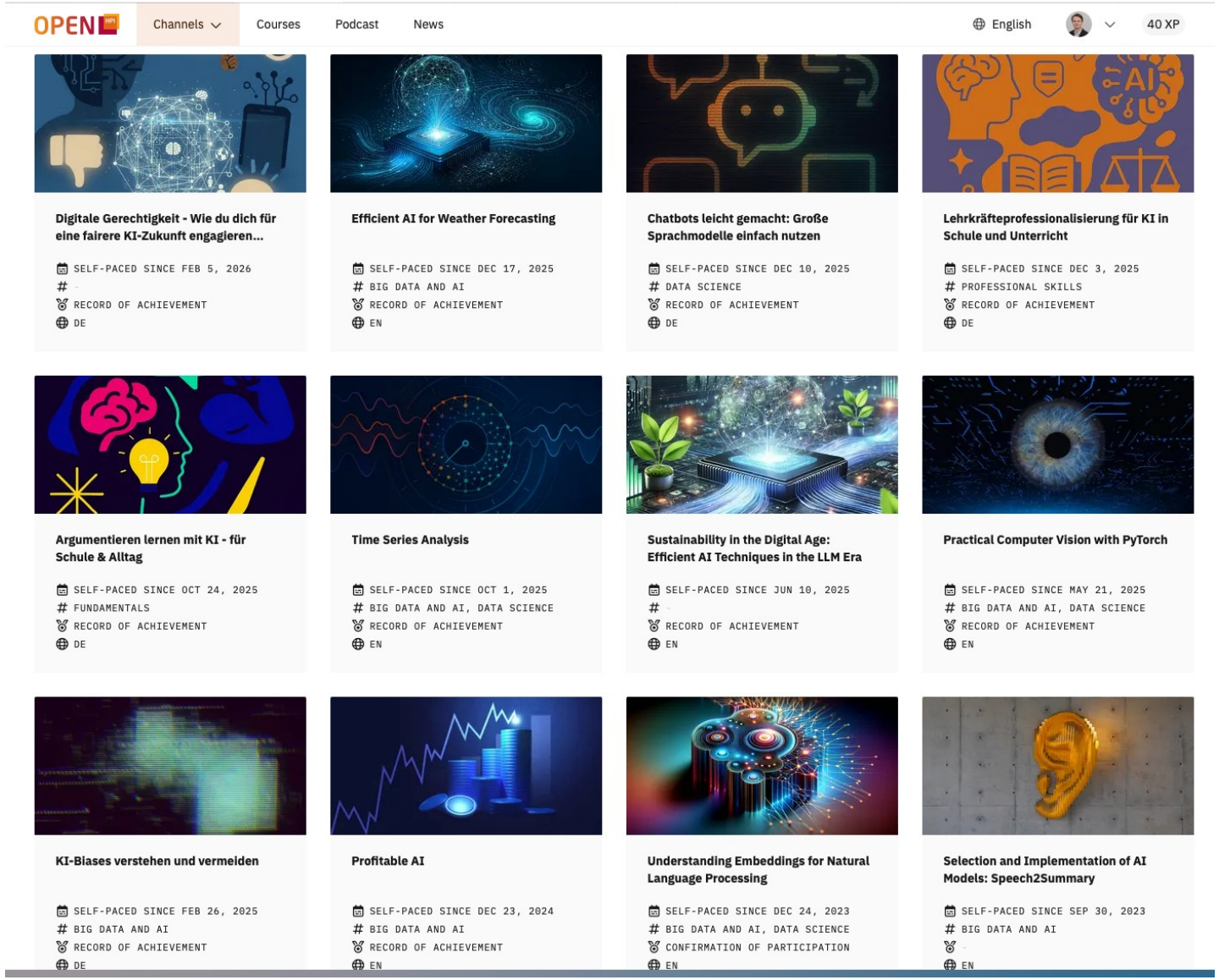


Effizientes Abfragen von Informationen – Teil IV

Di. 28. Apr., 15:00

Hasso -Plattner-Institut

Kostenlos


OPEN
Channels
Courses
Podcast
News
English
40 XP
Digitale Gerechtigkeit - Wie du dich für eine fairere KI-Zukunft engagieren...
SELF-PACED SINCE FEB 5, 2026
RECORD OF ACHIEVEMENT
DE
Efficient AI for Weather Forecasting
SELF-PACED SINCE DEC 17, 2025
BIG DATA AND AI
RECORD OF ACHIEVEMENT
EN
Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen
SELF-PACED SINCE DEC 10, 2025
DATA SCIENCE
RECORD OF ACHIEVEMENT
DE
Lehrkräfteprofessionalisierung für KI in Schule und Unterricht
SELF-PACED SINCE DEC 3, 2025
PROFESSIONAL SKILLS
RECORD OF ACHIEVEMENT
DE
Argumentieren lernen mit KI - für Schule & Alltag
SELF-PACED SINCE OCT 24, 2025
FUNDAMENTALS
RECORD OF ACHIEVEMENT
DE
Time Series Analysis
SELF-PACED SINCE OCT 1, 2025
BIG DATA AND AI, DATA SCIENCE
RECORD OF ACHIEVEMENT
EN
Sustainability in the Digital Age: Efficient AI Techniques in the LLM Era
SELF-PACED SINCE JUN 10, 2025
RECORD OF ACHIEVEMENT
EN
Practical Computer Vision with PyTorch
SELF-PACED SINCE MAY 21, 2025
BIG DATA AND AI, DATA SCIENCE
RECORD OF ACHIEVEMENT
EN
KI-Biases verstehen und vermeiden
SELF-PACED SINCE FEB 26, 2025
BIG DATA AND AI
RECORD OF ACHIEVEMENT
DE
Profitable AI
SELF-PACED SINCE DEC 23, 2024
BIG DATA AND AI
RECORD OF ACHIEVEMENT
EN
Understanding Embeddings for Natural Language Processing
SELF-PACED SINCE DEC 24, 2023
BIG DATA AND AI, DATA SCIENCE
CONFIRMATION OF PARTICIPATION
EN
Selection and Implementation of AI Models: Speech2Summary
SELF-PACED SINCE SEP 30, 2023
BIG DATA AND AI
EN

KI Service Zentrum
by Hasso-Plattner-Institut

HPI

kisz@hpi.de

hpi.de/kisz

Your opinion is important!

QR code for the feedback form



With funding from the:

Federal Ministry of Research, Technology and Space